

The empirical recurrence for OEIS A299064, recovered from its tail

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Abstract

OEIS A299064 records “Empirical recurrence of order 62” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 85$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 65$. Its last coefficient is $c_{62} = -536 \neq 0$, so no further tail satisfies anything shorter; any order-62 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299064 is “Number of $n \times 5$ 0..1 arrays with every element equal to 1, 2, 3, 4, 5 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 236, 7330, 196699, 5491159, 154373324, 4331069485, 121483918174, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Feb 01 2018 14:51:50, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 85$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 85$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 62: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 62.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 170$ exact terms from $n = 3$ on, it returns order 62 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	21	c_{17}	4312621043	c_{33}	4600751067	c_{49}	344076572
c_2	144	c_{18}	17315258677	c_{34}	47660279268	c_{50}	118622679
c_3	1283	c_{19}	23178114511	c_{35}	83282497862	c_{51}	66230547
c_4	6306	c_{20}	−1219387231	c_{36}	59685502999	c_{52}	29288925
c_5	3627	c_{21}	−67029755466	c_{37}	989350189	c_{53}	−18919032
c_6	−38070	c_{22}	−131367851151	c_{38}	−41097749155	c_{54}	−29734213
c_7	−446520	c_{23}	−93967103248	c_{39}	−42122712334	c_{55}	−13358380
c_8	−1448724	c_{24}	94537246700	c_{40}	−15654174235	c_{56}	1155549
c_9	1735061	c_{25}	313349625012	c_{41}	3826099397	c_{57}	3339029
c_{10}	13302981	c_{26}	301845300138	c_{42}	7561119032	c_{58}	1214443
c_{11}	22311501	c_{27}	−16296189478	c_{43}	7169531467	c_{59}	−32104
c_{12}	55236123	c_{28}	−329753021133	c_{44}	4355909868	c_{60}	−143124
c_{13}	−31384267	c_{29}	−298690917801	c_{45}	−1081169237	c_{61}	−27140
c_{14}	−661332822	c_{30}	−43691494758	c_{46}	−2657416556	c_{62}	−536
c_{15}	−1672632726	c_{31}	79656346547	c_{47}	−1198501545		
c_{16}	−1483981506	c_{32}	32931299710	c_{48}	95461181		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 65$, and it is the recurrence OEIS A299064 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{62} - c_1 x^{62-1} - \dots - c_{62}$. Its constant term is $-c_{62} = 536$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 62 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 62, so $\deg q = 62$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 62, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 62, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -536 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-62 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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